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THE MOTIF HUNT

Can you crack the code hidden in DNA?

Let's Talk Science x McGill CS Summer Camp Workshop
July 1st, 2026

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About Let's Talk Science

What We Do

Let's Talk Science is an award-winning, national, charitable organization focused on education and outreach to support youth development. We create and deliver unique learning programs and services that engage children, youth and educators in science, technology, engineering and mathematics (STEM).



- **Let's Talk Science at McGill University** is a local chapter of a national science outreach program. Together with our volunteers we work to bring hands-on activity kits to students from Preschool-CEGEP.



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What we're doing today

- **Background:** Genomics, machine learning, and deep learning
- **Live coding:** We train a neural network together on DNA data and find out the hidden motif
- **Your guess:** Submit your motif prediction on Slido
- **The reveal:** Did the AI find it?

What is DNA?

Every cell in your body contains DNA: a long sequence made of four letters: **A**, **C**, **G** and **T**. These letters are called **nucleotides**.

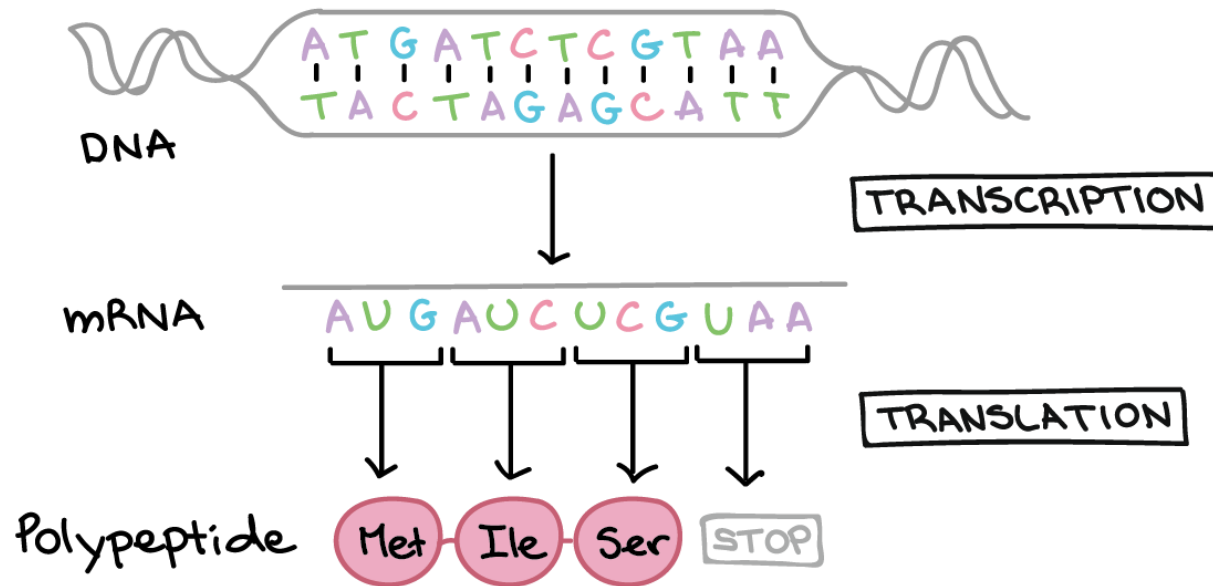
It stores the hereditary information an organism has, and is used to produce **proteins**. In humans, this sequence is about **3 billion letters long**.

This whole sequence is called **the genome**.



What is DNA?

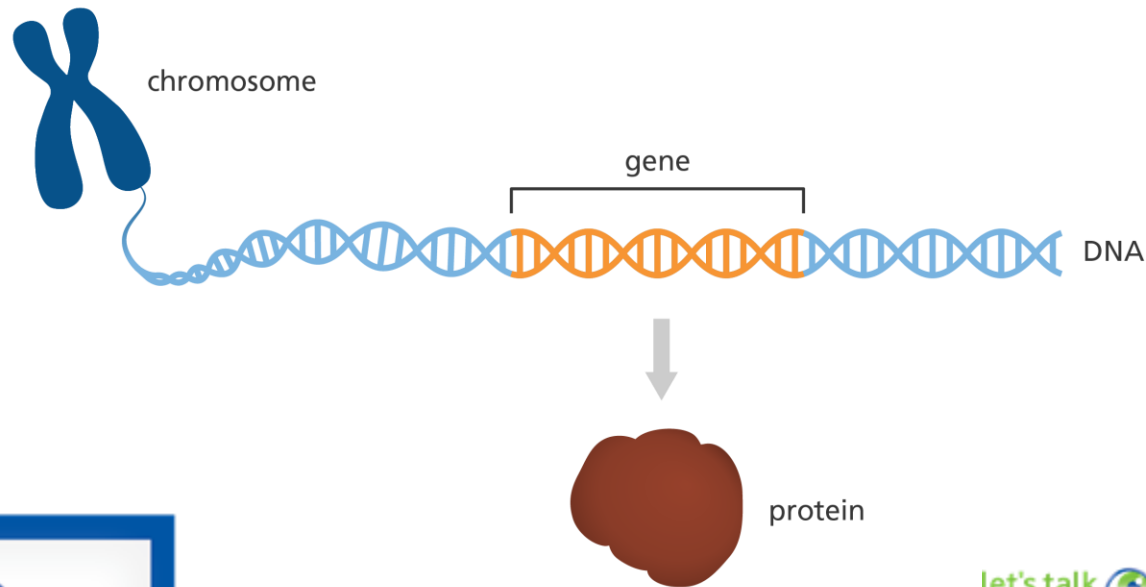
THE CENTRAL DOGMA



What is a gene?

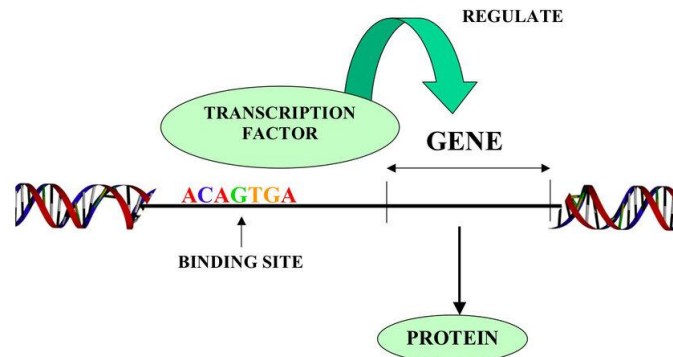
Genes are sections of the genome that contain instructions for making proteins.

~20,000 genes in the human genome.



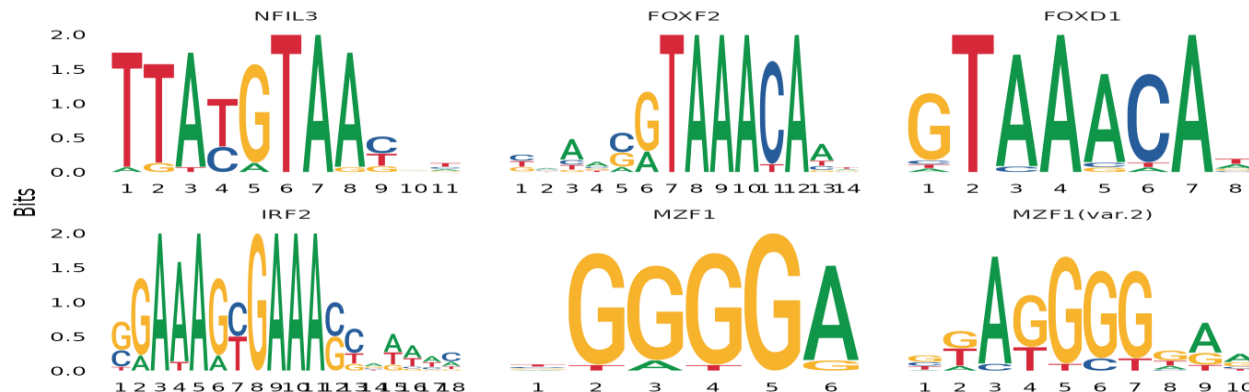
Transcription factors control gene activity

- Transcription factors are proteins that turn genes on or off. They do this by physically attaching to specific spots on the DNA, near the gene they control.
- The protein lands exactly where it recognizes a familiar sequence.



How do they know where to attach?

- Each transcription factor recognizes a specific short DNA sequence → a **motif**
- The transcription factor binds wherever this motif appears in the genome.



Why this matters?

- Knowing which motifs exist and where helps scientists understand how genes are regulated.
- This is very helpful to understand disease mechanisms (e.g., cancer) and better target them.

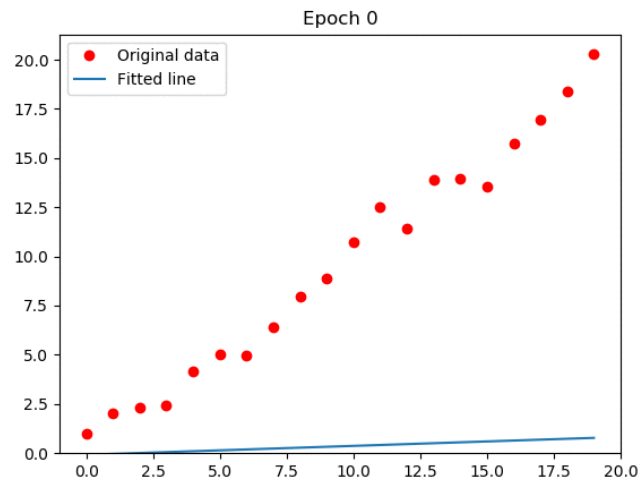
DNA is data!

- DNA is just a sequence of letters (A, C, G, T) repeated billions of times
- DNA can be studied the same way computer scientists study any large dataset



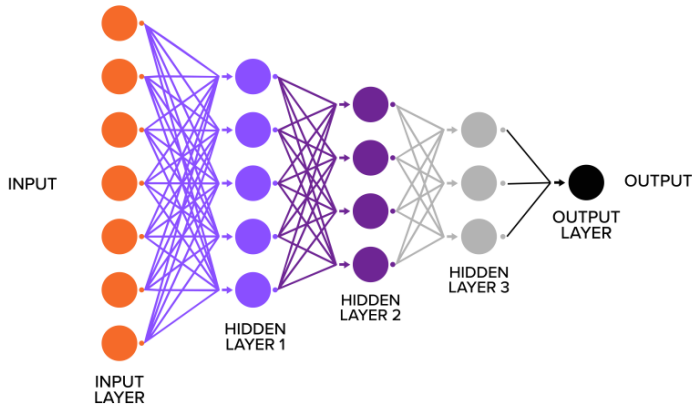
What is Machine Learning?

- Instead of writing explicit rules, you give the computer many examples with known answers. The computer finds patterns in the data on its own.



What is Deep Learning?

DEEP LEARNING WITH HIDDEN LAYERS



- Deep learning uses neural networks: layers of simple mathematical operations stacked on top of each other. Each layer transforms the data slightly.
- With enough layers, the network can represent very complex patterns.
- Deep learning can be used to solve many problems in life sciences!

Using Deep Learning to discover Motifs?

- We will use deep learning to **discover a hidden transcription-factor binding site (motif) in DNA!**
- We are looking for a specific motif. All we know is that it is composed of 12 nucleotides.
- Let's see if you get it right!



Before we start...

1. Go to: <https://github.com/NajiaB/The-Motif-Hunt>
2. Open the Google Colab link
3. Log into your Google Account
4. File -> Save a Copy in Drive -> Connect

Part 1 - The Dataset

Here's what we're working with:

- **2000** DNA sequences, each **50** bases long
- Each sequence is labelled **1** (protein bound) or **0** (protein didn't bind)
- The motif is hiding somewhere in the **positive** sequences

Part 1 - The Dataset

- Example of the first 4 sequences:

Sequence	bound
CCGAGGGCTATGGTTTGGAAGTTAGAACCCTGG GGCTTCTCGCGGA...	0
GAGTTTATATGGCGCGAGCCTAGTGGTTTTTGTA CTTGTTTGTCGC...	0
GATCAGTAGGGAAACAAACAGAGGGCCAGCC ACATCTAGCAGGTA...	0
GTCCACGACCGAACTCCACCTTGACCGCAGAG GTACCACCAGAGC...	1

Part 2 - How the AI Reads DNA

- Neural networks can't read letters!! We convert each base into a row of 0s and 1s.

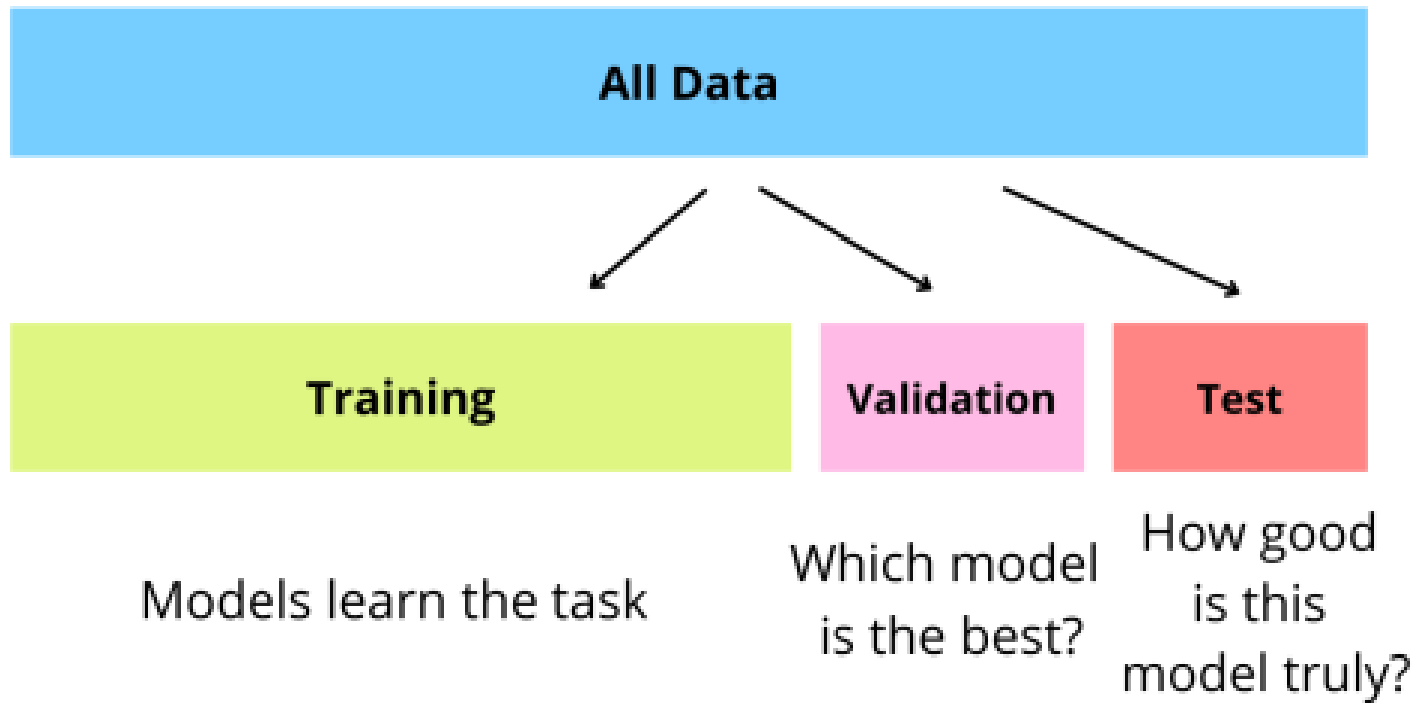
The diagram shows the generation of a 1D CNN feature map. The input is a 4x4 matrix of binary values (0s and 1s). The output is a 2x4 matrix of binary values. A red arrow points from the input matrix to the output feature map. A red bracket highlights the first row of the output feature map, which is the result of the 1D CNN operation.

	A	T	G	G	C	T	C	A	T
A	1	0	0	0	0	0	0	1	0
C	0	1	0	0	1	0	1	0	0
G	0	0	1	1	0	0	0	0	0
T	0	1	0	0	0	1	0	0	1

The output feature map is a 2x4 matrix of binary values:

	C	A	T
1	1	0	1
0	1	0	0

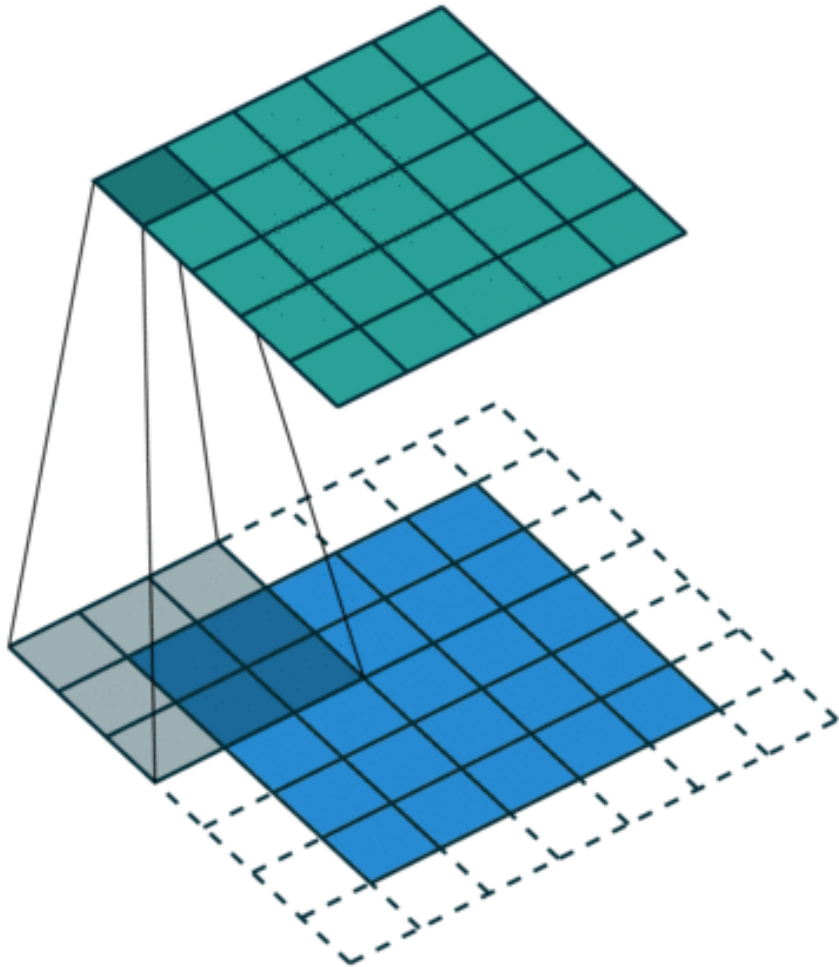
Part 2 - How the AI Reads DNA



Part 3 - The AI Model



Part 3 - The AI Model



- We're going to use a type of neural network called a Convolutional Neural Network (CNN)
- It uses **filters** to scan the sequences, like a magnifying glass sliding along the DNA, looking for a familiar shape.
- This architecture is widely used for images

Part 3 - The AI Model: Training

- We show the model thousands of labeled sequences, and it slowly adjusts its filters until it gets good at telling them apart.
- Each pass through the full dataset is called an epoch. Watch the loss go down and the accuracy go up.

Part 4 - Model Evaluation: Did our model actually learn something?

- We test the model on sequences it has never seen before. If it truly learned the pattern, it should do well on new data too!
- We can use a confusion matrix!
- What should our CM look like to confirm our model works?

		Actual Values	
		Positive	Negative
Predicted Values	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Part 5 - What did the model actually learn?

- A saliency map asks: if we slightly change a base, how much does the prediction change?
- They help unpack "black box" neural networks by visualizing exactly what features a model relies on to classify data: which nucleotides and at what position? => **Explainable AI**
- Tall bars = the AI is paying close attention to this base

What is your guess?

- After inspecting the attribution scores, what do you think the hidden motif is?
- Enter your guess here on Slido:



Motif reveal



Takeaways

- DNA can be represented as data and analyzed by computers
- Transcription factors control gene activity by recognizing short DNA patterns called motifs
- Neural networks find patterns by minimizing error
- With enough labeled examples, a model can discover biology it was never explicitly taught
- Saliency maps let us peek inside the model and ask: what did you actually learn?